Mathematics A

Unit 2 • Chapter OL-T45 Classified

Cross-session • chapter-organised candidate

18 classified items • 77 marks • 1 chapter(s)

Source-faithful practice with synchronized solutions

Dr Eslam Ahmed

Elite IGCSE Mathematics • eliteigcse.com

WhatsApp +20 112 000 9622

ELITE TEACHING EDITION - REVIEWED FOR WEBSITE USE

Contents

Unit 2 • Unit 2 • Chapter OL-T45 • 18 items • 77 marks

Chapter	Items	Marks	Starts
01. OL-T45 Volume & Surface Area	18	77	4

June 2026 is intentionally deferred; November 2025 remains provisional QP-only in this private candidate.

Volume & Surface Area

Unit 2 • 77 marks • 18 item(s)

June 2025 | 4MA1-1H Q7 | 3 marks
Recover a prism or cylinder dimension from volume

June 2025 | 4MA1-2H Q21 | 3 marks
Find surface area of a cuboid or simple prism

June 2024 | 4MA1-1H Q23 | 5 marks
Find volume or surface after removing a solid

November 2023 | 4MA1-1H Q21 | 5 marks
Find volume of a prism from its cross-section

November 2024 | 4MA1-1H Q5 | 4 marks
Convert terminal solid volume into metric capacity units

November 2024 | 4MA1-1H Q22 | 3 marks
Find volume or surface after removing a solid

November 2024 | 4MA1-2H Q6 | 3 marks
Find surface area of a cuboid or simple prism

June 2022 | 4MA1-1H Q7 | 4 marks
Find volume of a cuboid or cylinder

June 2022 | 4MA1-1H Q22 | 5 marks
Find volume of a joined composite solid

January 2020 | 4MA1-1H Q13 | 4 marks
Find volume of a cuboid or cylinder

January 2020 | 4MA1-2H Q26 | 6 marks
Find cone surface area or recover its sector

January 2021 | 4MA1-1H Q3 | 4 marks
Resolve transfer, capacity or rate involving prism volume

January 2021 | 4MA1-2H Q17 | 4 marks
Find surface area of a cuboid or simple prism

June 2021 | 4MA1-2H Q10 | 5 marks
Find volume of a prism from its cross-section

November 2021 | 4MA1-1H Q10 | 5 marks
Recover a prism or cylinder dimension from volume

... 3 more item(s) follow

Original question context and synchronized companion pages follow.

Terminal-demand ownership is preserved; prerequisite links remain in the internal ledger.

7 Here is a solid cylinder.

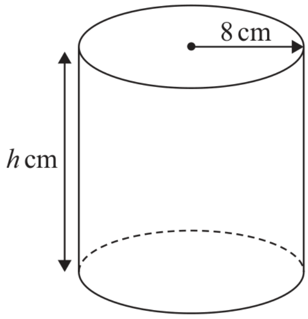


Diagram **NOT** accurately drawn

The radius of the cylinder is 8 cm
The height of the cylinder is h cm

The volume of the cylinder is 3892 cm^3

Work out the value of h
Give your answer correct to one decimal place.

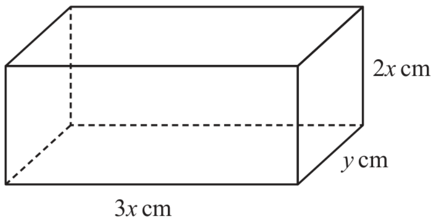
h = ...

(Total for Question 7 is 3 marks)

WORKING SPACE

21 The diagram shows a cuboid.

Diagram **NOT** accurately drawn



The cuboid measures 3x cm by 2x cm by y cm

The volume of the cuboid is 1014 cm³
The total surface area of the cuboid is A cm²

Show that $A = 12x^2 + \dfrac{1690}{x}$

You must show all the stages of your working.

(Total for Question 21 is 3 marks)

WORKING SPACE

Question 10
4MA1-1H Q23

MODULAR UNIT 2
5 marks

23 A solid shape is made by removing a hemisphere, shown shaded, from a cone as shown in the diagram.

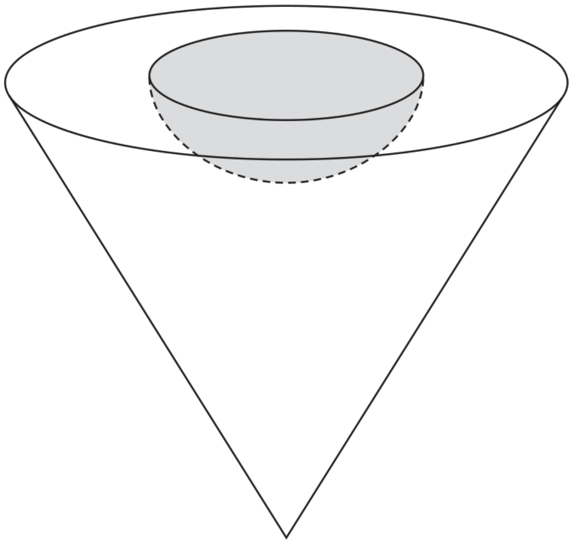


Diagram **NOT** accurately drawn

The radius of the hemisphere is $2x$ cm
The radius of the base of the cone is $5x$ cm
The vertical height of the cone is $6x$ cm

The volume of the solid shape is 6948π cm^3

Work out the **total** surface area of the solid hemisphere that has been removed from the cone.
Give your answer correct to the nearest integer.

.... cm^2

Working space for Question 10

Continue your method

UNIT 2
5 marks

WORKING SPACE

Question 16

4MA1-1H Q21 demand

MODULAR UNIT 2
5 marks

21 Here is a triangular prism *ABCDEF*

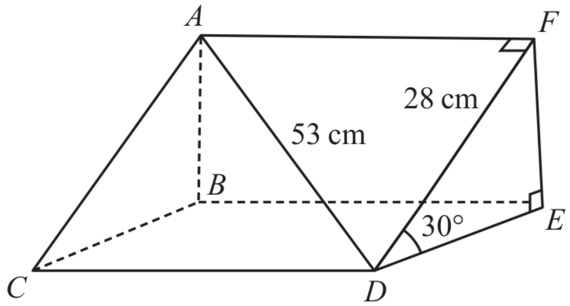


Diagram **NOT** accurately drawn

$AD = 53$ cm
$DF = 28$ cm
Angle $FDE = 30°$

Work out the volume of the triangular prism.
Give your answer correct to the nearest whole number.

.... cm^3

WORKING SPACE

Question 04
4MA1-1H Q5 M01

MODULAR UNIT 2
4 marks

5 The diagram shows a cylinder.

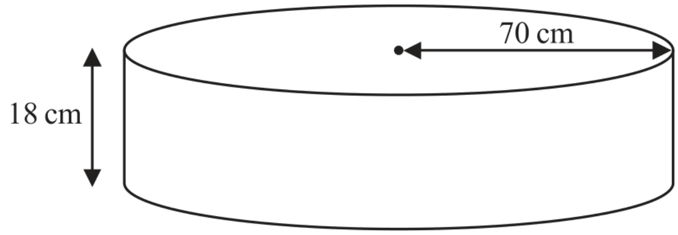


Diagram **NOT** accurately drawn

The radius of the cylinder is 70 cm
The height of the cylinder is 18 cm

Work out the volume of the cylinder.
Give your answer in litres correct to the nearest litre.

.... litres

WORKING SPACE

Question 14

4MA1-1H Q22 M01

MODULAR UNIT 2
3 marks

22 The diagram shows a bowl in the shape of a hemisphere.
The bowl is made from metal.

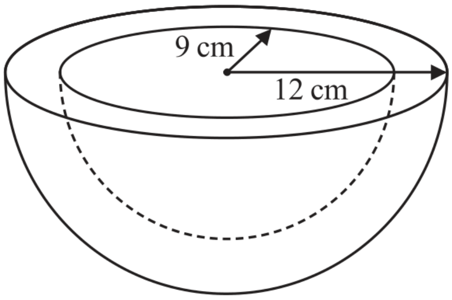


Diagram **NOT** accurately drawn

The outer radius of the bowl is 12 cm
The inner radius of the bowl is 9 cm

The thickness of the bowl is uniform.

Work out the volume of the metal.
Give your answer correct to the nearest whole number.

.... cm^3

WORKING SPACE

Question 20
4MA1-2H Q6 M01

MODULAR UNIT 2
3 marks

6 The diagram shows a solid triangular prism.

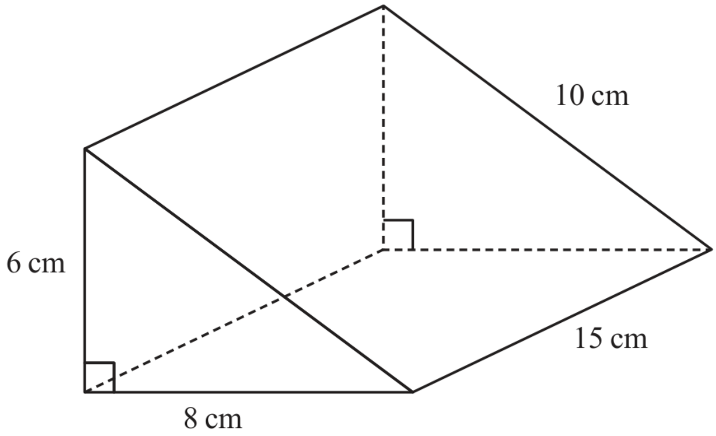


Diagram **NOT** accurately drawn

Work out the **total** surface area of the triangular prism.

.... cm^2

WORKING SPACE

Question 08
4MA1-1H Q7 Q7

MODULAR UNIT 2
4 marks

7 A cylinder is placed on the ground.

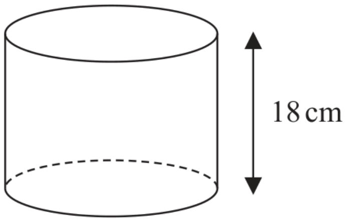


Diagram **NOT** accurately drawn

The height of the cylinder is 18 cm.

The force exerted by the cylinder on the ground is 72 newtons.
The pressure on the ground due to the cylinder is 1.4 newtons/cm^2

$$\boxed{\text{pressure} = \frac{\text{force}}{\text{area}}}$$

Work out the volume of the cylinder.
Give your answer correct to 3 significant figures.

.... cm^3

Working space for Question 08

Continue your method

UNIT 2
4 marks

WORKING SPACE

Question 18

4MA1-1H Q22 Q22

MODULAR UNIT 2
5 marks

22 A solid is made from a cone and a hemisphere.

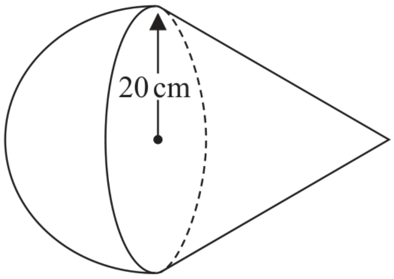


Diagram **NOT** accurately drawn

The circular plane face of the hemisphere coincides with the circular base of the cone.
The radius of the hemisphere and the radius of the circular base of the cone are both 20 cm.

The curved surface area of the cone is $580\pi \text{ cm}^2$

The volume of the solid is $k\pi \text{ cm}^3$

Work out the exact value of k

$k =$ ....

Working space for Question 18

Continue your method

UNIT 2
5 marks

WORKING SPACE

13 The diagram shows a solid cube.
The cube is placed on a table so that the whole of one face of the cube is in contact with the table.

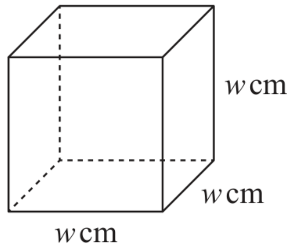


Diagram **NOT** accurately drawn

The cube exerts a force of 56 newtons on the table.
The pressure on the table due to the cube is 0.14 newtons/cm^2

$$\text{pressure} = \frac{\text{force}}{\text{area}}$$

Work out the volume of the cube.

.... cm^3

YOUR WORKING

26 Here is a sector, AOB, of a circle with centre O and angle $AOB = x°$

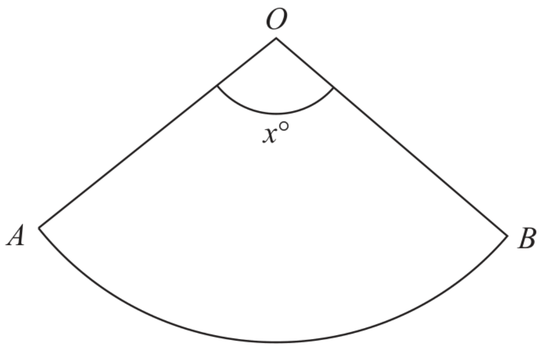


Diagram **NOT** accurately drawn

The sector can form the curved surface of a cone by joining OA to OB.

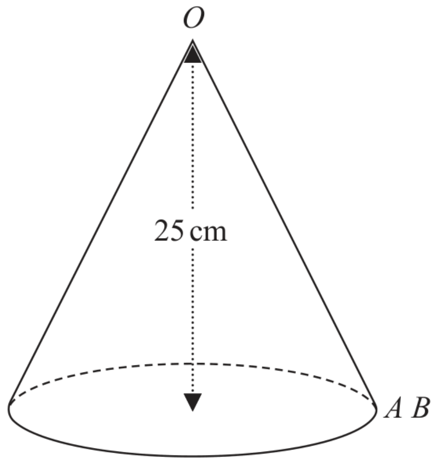


Diagram **NOT** accurately drawn

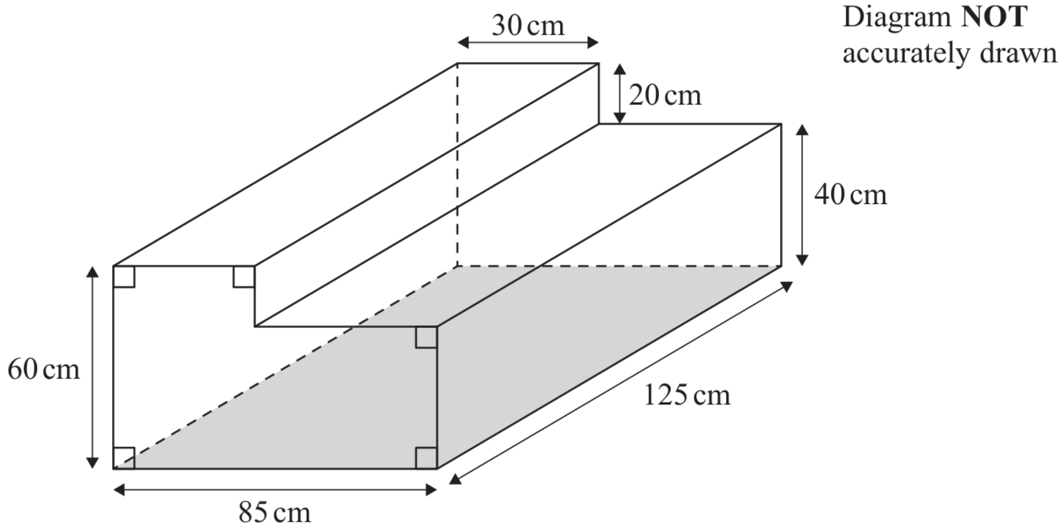
The height of the cone is 25 cm.
The volume of the cone is 1600 cm^3

Work out the value of x.
Give your answer correct to the nearest whole number.

$x =$ ....

YOUR WORKING

3 The diagram shows a container for water in the shape of a prism.



The rectangular base of the prism, shown shaded in the diagram, is horizontal.
The container is completely full of water.

Tuah is going to use a pump to empty the water from the container so that the volume of water in the container decreases at a constant rate.

The pump starts to empty water from the container at 10 30 and at 12 00 the water level in the container has dropped by 20 cm.

Find the time at which all the water has been pumped out of the container.

....

YOUR WORKING

17 The diagram shows a solid prism *ABCDEFGH*.

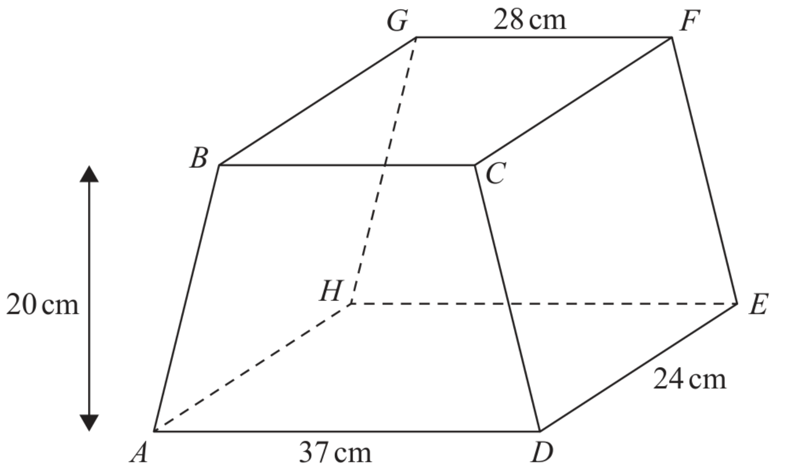


Diagram **NOT** accurately drawn

The trapezium *ABCD*, in which *AD* is parallel to *BC*, is a cross section of the prism.
The base *ADEH* of the prism is a horizontal plane.
ADEH and *BCFG* are rectangles.
The midpoint of *BC* is vertically above the midpoint of *AD* so that *BA* = *CD*.

AD = 37 cm   *GF* = 28 cm   *DE* = 24 cm

The perpendicular distance between edges *AD* and *BC* is 20 cm.

(a) Work out the total surface area of the prism.

.... cm^2

(4)

YOUR WORKING

10 Here is a triangular prism.

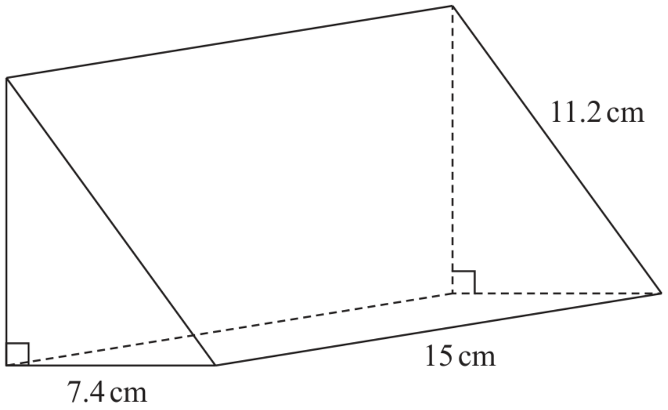


Diagram **NOT** accurately drawn

Work out the volume of the prism.
Give your answer correct to 3 significant figures.

.... cm^3

YOUR WORKING

10 A solid aluminium cylinder has radius 10 cm and height h cm.

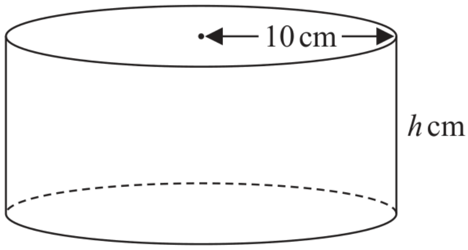


Diagram **NOT** accurately drawn

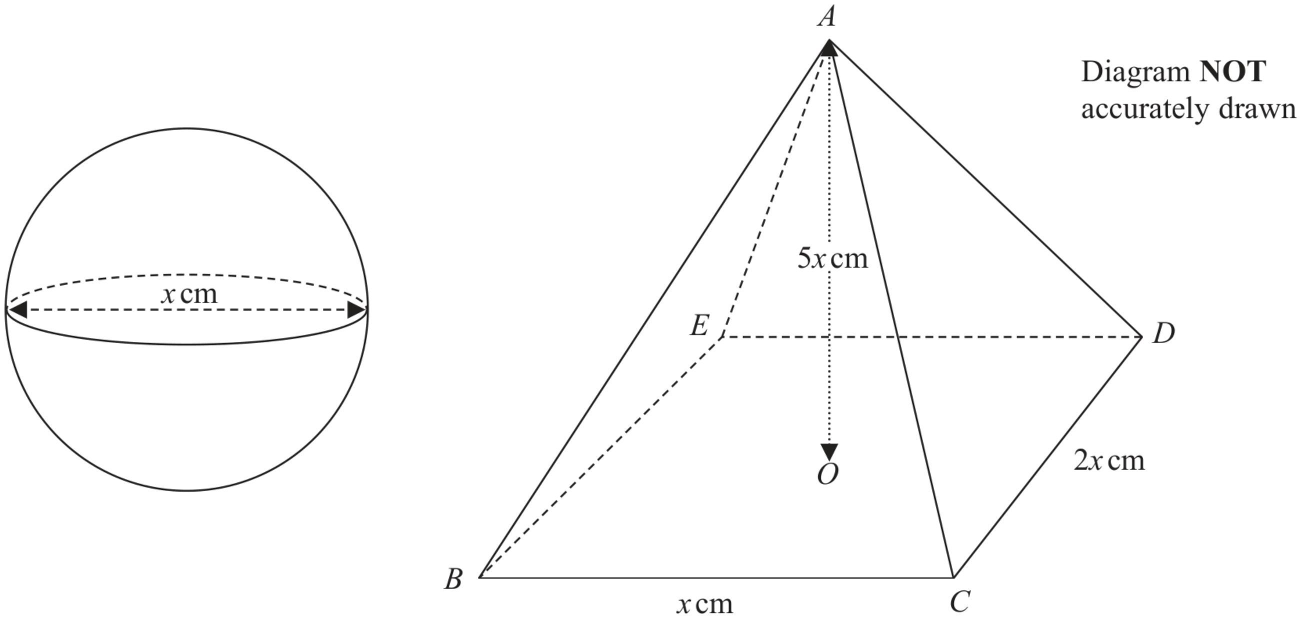
The mass of the cylinder is 5.4 kg.
The density of aluminium is 0.0027 kg/cm^3

Calculate the value of h.
Give your answer correct to one decimal place.

h = ...

YOUR WORKING

22 The diagram shows a sphere of diameter x cm and a pyramid $ABCDE$ with a horizontal rectangular base $BCDE$.



The vertex A of the pyramid is vertically above the centre O of the base so that $AB = AC = AD = AE$.

$BC = x$ cm, $CD = 2x$ cm and $AO = 5x$ cm.

The volume of the sphere is $288\pi \text{ cm}^3$

Calculate the total surface area of the pyramid.
Give your answer correct to the nearest cm^2

.... cm^2

YOUR WORKING

9 The diagram shows a solid cylinder with radius 3 m.

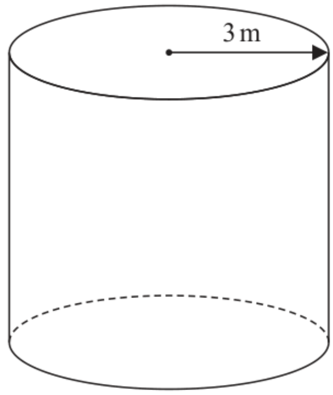


Diagram **NOT** accurately drawn

The volume of the cylinder is 72π m^3

Calculate the **total** surface area of the cylinder.
Give your answer correct to 3 significant figures.

.... m^2

(Total for Question 9 is 5 marks)

YOUR WORKING

21 Given that the surface area of a sphere is $49\pi\,\text{cm}^2$

find the volume of the sphere.
Give your answer correct to the nearest integer.

.... cm^3

(Total for Question 21 is 3 marks)

YOUR WORKING